

Dr. Tobias Weller

Lead AI Engineer

Singapore | Relocating from Berlin, Germany

Machine Learning Generative AI RAG Agentic AI Kubernetes AWS MLOps



Lead AI Engineer with 10+ years of experience building production AI systems at scale. Hands-on across the full AI lifecycle, from model development and platform design to deployment and operations. Built and operated AI platforms supporting 6,000+ users and processing 3.5M+ daily inference requests.

Professional Experience

German Federal Government — National Public Sector Organisation

Berlin, Germany

Lead AI Engineer

Dec. 2023 – Present

- Technical lead within an 18-engineer AI team, building and operating production AI systems used by 6,000+ employees.
- Built AI platforms processing 3.5M+ daily inference requests and established deployment and monitoring standards for large-scale AI services.
- Developed ML-driven decision systems processing 350,000+ documents daily, replacing rule-based workflows at scale.
- Built retrieval-augmented generation (RAG) systems and agent-based AI workflows integrating enterprise knowledge sources and external tools.
- Enabled 100+ departments to integrate AI capabilities into operational business processes.

University of Mannheim – Data and Web Science Group

Mannheim, Germany

Postdoctoral Researcher & Project Lead

Jan. 2021 – Nov. 2023

- Led AI research and development activities within a €5.7M international AI programme.
- Built machine learning, graph-based AI, and knowledge graph solutions with a focus on reliability and production readiness.
- Supervised a PhD researcher and coordinated technical deliverables across academic and industry partners.
- Focused on translating research prototypes into deployable real-world applications.

Karlsruhe Institute of Technology (KIT) – AIFB Institute

Karlsruhe, Germany

Research Scientist / PhD Candidate

Jun. 2015 – Dec. 2020

- Secured and led a €100k R&D project with DATEV, developing real-time anomaly detection systems for enterprise environments.
- Researched and developed machine learning methods with a focus on graph neural networks and uncertainty-aware AI.
- Built computer vision, OCR, and NLP solutions across research and industry collaborations.
- Delivered lectures and supervised student projects in machine learning and AI.

Education

Karlsruhe Institute of Technology (KIT)

Karlsruhe, Germany

PhD (Dr.-Ing.) in Machine Learning

Jun. 2015 – Dec. 2020

- Research focus: Graph Neural Networks and uncertainty-aware machine learning.

Karlsruhe Institute of Technology (KIT)

Karlsruhe, Germany

M.Sc. in Information Management and Engineering

Apr. 2013 – May 2015

- Graduated with High Distinction; focus on Machine Learning and Strategic Management.

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